# Erdős Problem 193 proof: notation and formula cheat sheet

Source of truth: `paper/erdos193.tex`, Sections 1–6. The proof ends before Section 7 (“Verification and evidence”).

## 1. Ambient conventions

| Notation | Meaning |
|---|---|
| $\mathbb{N}$ | Nonnegative integers $\{0, 1, 2, \ldots\}$ in this paper. |
| $\mathbb{Z}$, $\mathbb{Z}^2$, $\mathbb{Z}^3$ | Integers, planar integer lattice, and spatial integer lattice. |
| $\mathbb{F}_2$ | Field with two elements. Exponents reduced modulo $2$ encode parity. |
| $\lvert x\rvert$ | Absolute value. |
| $\lVert u\rVert_1$ | $\ell^1$ norm; for $u = (u_x, u_y)$, $\lVert u\rVert_1 = \lvert u_x\rvert + \lvert u_y\rvert$. |
| $f \circ g$ | Function composition with the right-hand map acting first: $(f \circ g)(z) = f(g(z))$. |
| $\nu_2(k)$ | Two-adic valuation: the exponent of the largest power of $2$ dividing nonzero integer $k$. The convention is $\nu_2(0) = \infty$ . |
| $q_j$ | Base-$4$ digit at place $j$; $j = 0$ is the least significant digit. |
| $x_j, y_j$ | Coordinate bits emitted at binary place $j$. |
| $u_x, u_y$ | Coordinates of a planar vector $u$. |

An infinite $S$-walk is a sequence $(P_a)_{a \geq 0} \subseteq \mathbb{Z}^3$ satisfying

$$P_{a+1} - P_a \in S \qquad (a \in \mathbb{N}),$$

for one fixed finite step set $S \subseteq \mathbb{Z}^3$.

## 2. Hilbert decoder

### State transformations

The decoder acts on a bit pair $(x, y) \in \{0, 1\}^2$ using

$$\begin{aligned} I(x,y) &= (x,y), & S(x,y) &= (y,x), \\ T(x,y) &= (1-y, 1-x), & C(x,y) &= (1-x, 1-y). \end{aligned}$$

Here $I$ is the identity, $S$ swaps the bits, $T$ swaps and complements them, and $C$ complements both. The state set is

$$K = \{I, S, T, C\} \cong \mathbb{F}_2^2,$$

with

$$S^2 = T^2 = C^2 = I, \qquad S \circ T = T \circ S = C.$$

Thus every $g \in K$ is an involution: $g^{-1} = g$.

During a decode, $g \in K$ is a **mutable local variable** for the current orientation state. It starts at

$$g = I.$$

After the last digit $q_0$ has been processed, define

$$\sigma(w) = \text{the resulting value of } g.$$

Thus $g$ denotes the changing intermediate state; $\sigma(w)$ denotes its final value. They are related, but they are not interchangeable during the scan.

## Digit data

The four child positions, in U-shaped traversal order, are

$$c_0 = (0,0), \qquad c_1 = (0,1), \qquad c_2 = (1,1), \qquad c_3 = (1,0).$$

The symbol $r_q$ denotes the transition map selected by digit $q$; it is not a numeric sequence. The lookup table is

$$r_0 = S, \qquad r_1 = I, \qquad r_2 = I, \qquad r_3 = T.$$

In

$$g \longleftarrow g \circ r_q,$$

"update" means **replace the current orientation map $g$ by the composed map $g \circ r_q$** after digit $q$ emits its coordinate bits. The subscripted map $r_q \in K$ is unrelated to the scalar integer $r$ used later in the collinearity proof.

For a length-$\ell$ base-$4$ word

$$w = q_{\ell-1} \cdots q_1 q_0,$$

the decoder reads digits from most significant to least significant, begins in state $I$, and at digit $q_j$ performs

$$(x_j, y_j) = g(c_{q_j}), \qquad g \longleftarrow g \circ r_{q_j}.$$

Reading order and output place run oppositely: $q_{\ell-1}$ is read first, but the pair emitted by $q_j$ occupies binary place $j$.

The finite-order Hilbert point is

$$H_\ell(w) = \left( \sum_{j=0}^{\ell-1} x_j 2^j, \sum_{j=0}^{\ell-1} y_j 2^j \right).$$

The map $H_\ell$ is a bijection from length-$\ell$ base-$4$ words to $\{0, \ldots, 2^\ell - 1\}^2$. Consecutive words represent planar lattice neighbors.

The finite-order induction also uses the convention that the order-$\ell$ route starts at $(0,0)$ and ends at $(2^\ell - 1, 0)$.

Writing $N = 2^\ell$, bitwise action on an $N \times N$ child square is

$$S(x,y) = (y,x), \qquad T(x,y) = (N-1-y, N-1-x).$$

The four transformed order-$\ell$ copies have start–end pairs

$$\begin{aligned}
(0,0) &\longrightarrow (0, N-1), \\
(0,N) &\longrightarrow (N-1, N), \\
(N,N) &\longrightarrow (2N-1, N), \\
(2N-1, N-1) &\longrightarrow (2N-1, 0),
\end{aligned}$$

and junction displacements $(0,1)$, $(1,0)$, and $(0,-1)$. A suffix decoded from state $g$ rather than $I$ has state $g \circ h$ whenever the $I$-started suffix has state $h$, and its emitted route is the bitwise $g$-transform of the $I$-started route.

## Padding, infinite path, and terminal state

Two leading zero digits do not change the point or terminal state:

$$H_{\ell+2}(00w) = H_\ell(w), \qquad \sigma(00w) = \sigma(w).$$

For $n \in \mathbb{N}$, use any sufficiently long, even-length, zero-padded base-$4$ representation of $n$:

- $H(n) \in \mathbb{N}^2$ is its finite-order Hilbert point;
- $\sigma(n) \in K$ is the state left after decoding it.

The same symbol $\sigma(w)$ denotes the terminal state of a finite word. Context distinguishes word input from integer input.

The infinite path $H : \mathbb{N} \to \mathbb{N}^2$ is injective and has unit steps:

$$\|H(n+1) - H(n)\|_1 = 1.$$

Only digits $0$ and $3$ change the state. If $N_0(w)$ and $N_3(w)$ count those digits, then

$$\sigma(w) = S^{N_0(w) \bmod 2} T^{N_3(w) \bmod 2}.$$

Thus $\sigma$ records exactly the parity pair $(N_0 \bmod 2, N_3 \bmod 2)$.

If digit $q$ leaves outgoing state $h$, its incoming state and emitted pair can be recovered backward:

$$\text{incoming state} = h \circ r_q,$$
$$(h \circ r_q)(c_q) = h(r_q(c_q)) = h(c_q),$$

because $r_q(c_q) = c_q$.

## 3. Two-adic chord invariant

For a nonzero planar chord $u = (u_x, u_y) \in \mathbb{Z}^2 \setminus \{0\}$, define

$$p(u) = \min\{\nu_2(u_x), \nu_2(u_y)\},$$
$$\epsilon(u) = \begin{cases} 1, & \nu_2(u_x) = \nu_2(u_y), \\ 0, & \nu_2(u_x) \neq \nu_2(u_y), \end{cases}$$

and

$$V(u) = 2p(u) + \epsilon(u).$$

Interpretation: $p(u)$ is the largest common binary scale of the two coordinates. The bit $\epsilon(u)$ says whether both normalized coordinates first become odd at that scale.

For every positive integer $k$,

$$V(ku) = V(u) + 2\nu_2(k).$$

The factor $2$ appears because each binary scale has two possible $V$-values: one coordinate first appears ($2j$), or both do ($2j+1$).

### First-mismatch convention

For equal-length, even-padded words $q_{\ell-1} \cdots q_0$ and $q'_{\ell-1} \cdots q'_0$, define the first mismatch **from the low end** by

$$j = \min\{k \geq 0 : q_k \neq q'_k\}.$$

Digits $q_0, \ldots, q_{j-1}$ form the common low suffix. If the two words have equal terminal state, their Hilbert coordinates agree below binary place $j$, and at place $j$:

- exactly one coordinate bit differs if $q_j - q'_j$ is odd;
- both coordinate bits differ if $q_j - q'_j \equiv 2 \pmod 4$.

In the pair-law proof, the local names are

$$d = q_j, \qquad e = q'_j,$$

and, for \(m,

$$n - m = 4^j M, \qquad M \equiv e - d \pmod 4.$$

A differing coordinate has the form

$$\pm 2^j + 2^{j+1} L = 2^j(\pm 1 + 2L),$$

where $L \in \mathbb{Z}$ and the parenthesized factor is odd.

### Pair law

For distinct indices with the same terminal state,

$$\sigma(m) = \sigma(n) \quad \Longrightarrow \quad V(H(n) - H(m)) = \nu_2(|n - m|).$$

This is the proof's central invariant.

## 4. Bounded-gap selector and lifted walk

For block index $a \geq 0$, let $g = \sigma(a)$ and choose a two-digit base-$4$ suffix through the offset map $\rho : K \to \{1, 3, 5, 13\}$:

| **Prefix state** $g$ | **Chosen suffix** | **Decimal offset** $\rho(g)$ | **Full state chain** |
|---|---|---|---|
| $I$ | $11_4$ | $5$ | $I \circ I \circ I = I$ |
| $S$ | $01_4$ | $1$ | $I \circ S \circ S = I$ |
| $T$ | $31_4$ | $13$ | $I \circ T \circ T = I$ |
| $C$ | $03_4$ | $3$ | $I \circ C \circ C = I$ |

Define the selected index

$$n_a = 16a + \rho(\sigma(a)).$$

Multiplication by $16 = 4^2$ appends the chosen two base-$4$ digits. Since every $g \in K$ is its own inverse,

$$\sigma(n_a) = I \circ g \circ g = I.$$

The selected gaps satisfy

$$n_{a+1} - n_a = 16 + \rho(\sigma(a+1)) - \rho(\sigma(a)),$$
$$4 \leq n_{a+1} - n_a \leq 28.$$

Two lift notations are used:

$$Q(n) = (H(n), n)$$

for the generic lift of an index, and

$$P_a = Q(n_a) = (H(n_a), n_a) \in \mathbb{Z}^3$$

for the selected walk. Every $P_a$ has terminal state $I$ through its index $n_a$.

## 5. Collinearity contradiction

The same-state lift theorem says: if $E \subseteq \mathbb{N}$ and $\sigma$ is constant on $E$, then

$$\{Q(n) : n \in E\}$$

contains no three collinear points.

For a hypothetical collinear triple with \(a in $E$, define adjacent index gaps and planar chords by

$$A = b - a, \qquad B = c - b,$$
$$X = H(b) - H(a), \qquad Y = H(c) - H(b).$$

The corresponding spatial displacements are $(X, A)$ and $(Y, B)$. Because $A, B > 0$, collinearity is equivalent to

$$BX = AY.$$

Factor the gaps as

$$A = gr, \qquad B = gs,$$
$$g = \gcd(A, B), \qquad \gcd(r, s) = 1.$$

Cancelling $g$ gives $sX = rY$. Coordinatewise divisibility yields an integer vector $W \in \mathbb{Z}^2 \setminus \{0\}$ with

$$X = rW, \qquad Y = sW.$$

Applying the pair law and scaling law to the adjacent pairs gives

$$\nu_2(g) + \nu_2(r) = V(X) = V(W) + 2\nu_2(r),$$

$$\nu_2(g) + \nu_2(s) = V(Y) = V(W) + 2\nu_2(s).$$

Hence

$$\nu_2(r) = \nu_2(s) = 0, \qquad V(W) = \nu_2(g),$$

so coprime $r$ and $s$ are both odd.

For the endpoint pair,

$$c - a = g(r+s), \qquad H(c) - H(a) = (r+s)W.$$

Set

$$t = \nu_2(r+s).$$

The pair and scaling laws give

$$\nu_2(g) + t = V((r+s)W) = V(W) + 2t = \nu_2(g) + 2t.$$

Therefore $t = 0$, which says $r+s$ is odd. But $r$ and $s$ are both odd, so $r+s$ is even: contradiction.

## 6. Finite step menu

Write one successive spatial displacement as

$$P_{a+1} - P_a = (d_x, d_y, d_z).$$

Its height increment is

$$d_z = n_{a+1} - n_a, \qquad 4 \le d_z \le 28.$$

Unit Hilbert steps and the triangle inequality give

$$|d_x| + |d_y| = \|H(n_{a+1}) - H(n_a)\|_1 \le n_{a+1} - n_a = d_z.$$

A fixed finite menu containing every realized step is therefore

$$S = \{(d_x, d_y, d_z) \in \mathbb{Z}^3 : 4 \le d_z \le 28,\ |d_x| + |d_y| \le d_z\}.$$

This $S$ is a convenient finite superset, not the minimal realized menu.

## 7. Letter index

### Global or recurring symbols

| Symbol | Type | Role |
|---|---|---|
| $S$ | transformation, then set | Decoder swap $S(x,y) = (y,x)$ in Sections 2–4; finite spatial step set in the theorem and Section 5. See the collision warning below. |
| $P_a$ | $\mathbb{Z}^3$ | $a$th vertex of the selected infinite walk. |
| $H_\ell$ | finite words $\to \mathbb{N}^2$ | Finite-order Hilbert decoder. |
| $H$ | $\mathbb{N} \to \mathbb{N}^2$ | Nested infinite Hilbert path. |
| $\sigma$ | words or $\mathbb{N} \to K$ | Terminal value of the mutable decoder state $g$ after the last digit. |
| $K$ | four-element set/group | Decoder state space $\{I, S, T, C\}$. |
| $I, S, T, C$ | maps on $\{0,1\}^2$ | Four decoder transformations. |
| $c_q$ | $\{0,1\}^2$ | Child position associated with digit $q \in \{0,1,2,3\}$. |
| $r_q$ | $K$ | Digit-indexed transition map; after digit $q$, replace the current state $g$ by $g \circ r_q$. Not the scalar $r$ in the triple argument. |

| Symbol | Type | Role |
|---|---|---|
| $w$ | base-4 word | Finite decoder input. |
| $\ell$ | nonnegative integer | Word length / finite Hilbert order. |
| $q_j, q'_j$ | base-4 digits | Digits of two compared words at place $j$. |
| $x_j, y_j$ | bits | Coordinate bits emitted at place $j$. |
| $N_0(w), N_3(w)$ | nonnegative integers | Counts of digits $0$ and $3$ in $w$. |
| $u = (u_x, u_y)$ | nonzero $\mathbb{Z}^2$ vector | Generic planar chord. |
| $p(u)$ | nonnegative integer | Common two-adic coordinate scale. |
| $\epsilon(u)$ | $\{0, 1\}$ | Whether the coordinate valuations are equal. |
| $V(u)$ | nonnegative integer | Two-adic planar chord invariant. |
| $\rho$ | $K \to \{1, 3, 5, 13\}$ | State-dependent selector offset. |
| $n_a$ | nonnegative integer | Selected Hilbert index in block $a$. |
| $Q(n)$ | $\mathbb{Z}^3$ | Generic lift $(H(n), n)$. |
| $E$ | subset of $\mathbb{N}$ | Arbitrary same-terminal-state index set. |
| $d_x, d_y, d_z$ | integers | Coordinates of one successive step of $P$. |

## Proof-local symbols

| Symbol | Scope | Meaning |
|---|---|---|
| $g, h$ | Decoder arguments | $g$ is the mutable current state, initialized at $I$, whose final value is $\sigma(w)$; $h$ is an outgoing or common state used in reverse decoding. |
| $d, e$ | First-mismatch lemma | Unequal digits $q_j, q'_j$. |
| $j$ | First-mismatch argument | Least digit place where two base-4 words differ. |
| $M$ | Pair-law proof | Integer remaining after $n - m = 4^j M$. |
| $L$ | Pair-law proof | Integer collecting higher binary-place contributions. |
| $k$ | Scaling lemma | Positive integer scalar; in the mismatch definition, a dummy digit-place index. |
| $a$ | Selector sections | Block number indexing $n_a$ and $P_a$. |
| $a, b, c$ | Triple-free theorem | Three ordered indices in $E$; this reuses $a$ with a different local role. |
| $A, B$ | Triple-free theorem | Positive adjacent gaps $b - a$ and $c - b$. |
| $X, Y$ | Triple-free theorem | Adjacent planar chords. |
| $g$ | Triple-free theorem | $\gcd(A, B)$; this reuses $g$ with a different local role. |
| $r, s$ | Triple-free theorem | Coprime reduced gap factors: $A = gr$, $B = gs$. |
| $W$ | Triple-free theorem | Common nonzero primitive-direction multiple with $X = rW$, $Y = sW$. |
| $t$ | Triple-free theorem | $\nu_2(r + s)$. |

## 8. Collision and reading warnings

1. $S$ **is overloaded.** In the decoder, $S$ is the swap transformation. In the theorem and final section, $S \subseteq \mathbb{Z}^3$ is the finite step menu. Type and section determine the meaning.
2. $g$ **is local and overloaded.** In the decoder, $g \in K$ is the mutable current state: it starts at $I$, and its final value is $\sigma(w)$. In the collinearity proof, the unrelated integer $g$ means $\gcd(A, B)$.
3. $r_q$ **and** $r$ **are different objects.** The subscripted $r_q \in K$ is the transition map selected by base-$4$ digit $q$. The unsubscripted $r \in \mathbb{N}$ is the coprime reduced factor in $A = gr$.
4. $a$ **changes role.** It is the block/walk index in $n_a, P_a$, but one of three arbitrary ordered indices \(a inside the same-state lift theorem.
5. $\sigma$ **is not a coordinate.** It is the accumulated terminal transformation left by decoding.
6. **Most-significant-first reading, least-significant place labels.** The decoder reads $q_{\ell-1}, \ldots, q_0$, while $q_j$ emits at binary place $j$.
7. **"First mismatch" means from the low end.** It is the least $j$, not the first digit encountered by the forward decoder.
8. $H(n)$ **is planar;** $Q(n)$ **and** $P_a$ **are spatial.** The third coordinate is the Hilbert index itself.
9. **The menu is not the realized set.** The displayed finite $S$ contains all possible successive displacements allowed by the proved bounds; it need not be minimal.